

Zero-Shot Confidence Estimation for Small LLMs: When Supervised Baselines Aren't Worth Training

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Abstract

Local-to-cloud LLM routing decides whether a cheap local model can answer a query or whether to escalate to an expensive cloud model. The dominant approach trains a supervised classifier on labeled routing examples. We ask whether that training is necessary.

We compare six zero-shot confidence signals against RouteLLM-style supervised baselines across three local model families (Qwen-2.5-7B, Llama-3.1-8B, Mistral-7B-Instruct) and two datasets (MMLU-Pro, TriviaQA; $\sim 3,000$ evaluation queries per model). We find that (1) average token log-probability, a zero-shot signal requiring no training data, matches or exceeds supervised baselines in-distribution (AUROC 0.650–0.714 vs. 0.644–0.676) and substantially outperforms them out-of-distribution (0.717–0.833 vs. 0.512–0.564); (2) supervised routing transfers across model families but collapses on a new dataset; (3) naive multi-signal fusion hurts rather than helps; and (4) signal quality correlates with RLHF calibration training across model families. We provide a cold-start deployment guide and release all code, data, and experiment logs.

1 Introduction

Running a small local model (7–8B parameters) for most queries and escalating to a cloud model only when the local model is likely to fail is an increasingly common architecture for cost-sensitive LLM deployments. The routing decision—local or cloud?—determines both cost and quality. A growing body of work addresses this problem, but the field has converged on supervised approaches without adequately testing whether the supervision is necessary.

1.1 Supervised Routing

RouteLLM [Ong et al., 2024] established the dominant paradigm: train a classifier on labeled (query, model-was-sufficient) pairs. Subsequent work extends this to contextual bandits under budget constraints [Chen et al., 2025], bandit feedback with regret guarantees [Soare et al., 2025], dueling feedback [Li et al., 2025], and non-stationary settings where pricing and quality shift over time [Taberner-Miller et al., 2026]. Gomes et al. [2025] survey routing strategies more broadly.

All of these approaches require training data—labeled examples, preference pairs, or online feedback. For a new deployment, this means generating answers from both local and cloud models on a representative corpus, labeling correctness, and fitting a routing model. In our experiments, this costs approximately \$25 and 2 hours per local model.

1.2 Zero-Shot Confidence Signals

An alternative extracts confidence signals directly from the local model’s inference, with no training data.

Token-level calibration. Kadavath et al. [2022] showed that RLHF’d models produce calibrated token probabilities that correlate with correctness. Guo et al. [2017] established calibration theory for deep

networks. [Huang et al. \[2024\]](#) demonstrated that log-probabilities remain reliable plausibility estimates in instruction-tuned models.

Self-assessment. [Kadavath et al. \[2022\]](#) tested P(True) prompting for self-evaluation. [Ren et al. \[2023\]](#) reformulated generation into token-level self-evaluation, improving selective generation on TruthfulQA. [Chen et al. \[2024\]](#) found self-reported confidence achieves better calibration than self-consistency at $5\times$ lower cost.

Multi-signal uncertainty. UQLM [[Lin et al., 2025](#)] and MUSE [[Geng et al., 2025](#)] combine multiple uncertainty signals into unified frameworks. [Mukkunth et al. \[2025\]](#) combine semantic alignment, internal convergence, and learned confidence for pre-generation hallucination routing. [Wang et al. \[2022\]](#) introduced self-consistency—sampling multiple reasoning paths and measuring agreement—as a confidence proxy.

1.3 Uncertainty-Based Routing

The closest work to ours is [Chuang et al. \[2025\]](#), who benchmark uncertainty-based SLM-to-LLM routing across 1,500+ settings. They find that uncertainty distributions depend more on the SLM and quantification method than on downstream data, and propose a calibration data construction pipeline for generalization. [Zhang et al. \[2025\]](#) propose a confidence-driven router for edge-cloud deployment, evaluating on MT-Bench, GSM8K, and MMLU.

1.4 The Gap We Address

Despite this active literature, a specific comparison is missing. Existing routing papers assume supervision is available and do not benchmark against zero-shot alternatives. Existing confidence papers study signal quality in isolation without comparing to supervised routing baselines. [Chuang et al. \[2025\]](#) benchmark uncertainty methods against each other, not against supervised routers. [Lugoloobi et al. \[2026\]](#) demonstrate that linear probes on pre-generation activations can predict success and guide routing, but require access to model internals and per-model probe training; they do not compare against supervised baselines or test cross-dataset transfer. No published work evaluates zero-shot confidence signals against supervised routing baselines across multiple model families *and* multiple datasets.

We fill this gap and find that the supervised approach is not worth the investment for most deployments.

1.5 Contributions

1. A cross-model, cross-dataset comparison of six zero-shot confidence signals against supervised baselines (3 models $\times$ 2 datasets $\times$ $\sim$ 1,000 queries each), with bootstrap CIs and paired significance tests.
2. The finding that supervised routing is dataset-specific while zero-shot log-probability transfers across both models and datasets (§4.2).
3. A learning curve showing supervised routing converges at $\sim$ 250 examples but never exceeds the zero-shot baseline (§4.3).
4. Evidence that signal quality correlates with RLHF calibration training, with logprob and self-assessment complementary across the model-family axis (§5.1).
5. A cold-start deployment guide (§6) and a retrieval-conditional self-assessment design principle (§4.6).

All experiments are reproducible on a single laptop with Ollama and AWS Bedrock. Total cloud cost: \$123.

2 Signals and Baselines

2.1 Zero-Shot Confidence Signals

We evaluate six signals computed at inference time without training data.

Log-probability (logprob). Average per-token log-probability of the local model’s generated answer, mapped to $[0, 1]$ via $\sigma(\bar{\ell} \cdot 2 + 3)$ where $\bar{\ell}$ is the mean token log-prob. Higher values indicate greater token-level confidence [Kadavath et al., 2022].

Self-assessment (GSA). A single-token YES/NO probe: the model is asked “Are you confident you can answer this question correctly?” and the YES-token probability is extracted from the first position’s logprobs. We test a bare variant and a retrieval-conditional variant (v3) that injects strong retrieved hits when available but falls back to a byte-identical bare prompt otherwise, so the model cannot distinguish “retrieval found nothing” from “retrieval was never attempted.”

Self-consistency (SC). Jaccard token overlap between two independent generations at temperature 0.0 and 0.7 [Wang et al., 2022].

Knowledge similarity (KS). Maximum cosine similarity between the query embedding and top-5 retrieved KB entries via FAISS inner-product on L2-normalized bge-large-en-v1.5 embeddings (1024-dim).

Query classification (QC). Keyword heuristic: factual $\rightarrow$ 0.7, real-time $\rightarrow$ 0.2, creative $\rightarrow$ 0.4, default $\rightarrow$ 0.5.

Energy scorer (ES). Logistic regression on query embeddings trained on the agent’s pass/fail history. Disabled in all experiments (requires 50+ labeled examples).

2.2 Supervised Baselines

Following RouteLLM [Ong et al., 2024], we train two logistic regression classifiers (LogisticRegressionCV, 5-fold) on a disjoint 1,000-query training corpus per model:

- **RouteLLM-nm:** query embedding $\rightarrow P(\text{local sufficient})$
- **RouteLLM-pks:** [query embedding||knowledge similarity] $\rightarrow P(\text{local sufficient})$

3 Experimental Setup

3.1 Models

Table 1: Local models evaluated.

Model	Params	RLHF emphasis
Qwen-2.5-7B	7.6B	Calibration-aware
Llama-3.1-8B-Instruct	8.0B	Helpfulness / harmlessness
Mistral-7B-Instruct	7.2B	Mixed

Cloud model: Claude Sonnet 4.5 (AWS Bedrock). Judge: Claude Opus 4.5.

3.2 Datasets

MMLU-Pro (primary): 10-option MCQ, 14 categories. KB: 995 seeded entries. Eval: 931–999 queries per model.

TriviaQA (secondary): open-ended short-answer. KB: 200 seeded entries. Eval: 500 queries per model.

3.3 Labeling

MMLU-Pro uses regex letter extraction with LLM-judge fallback. TriviaQA uses standard substring matching.

Table 2: Judge fallback rates on MMLU-Pro.

Model	Judge calls / total	Rate
Qwen-2.5-7B	688 / 931	74%
Llama-3.1-8B	755 / 997	76%
Mistral-7B	972 / 999	97%

Mistral’s 97% judge rate reflects poorly formatted responses; its AUROC is more sensitive to judge noise. Label noise bound: 13% inter-judge disagreement (Opus vs. Sonnet audit on 100 rows). Differential comparisons are robust to symmetric noise.

3.4 Evaluation Protocol

AUROC with 1,000-sample bootstrap CIs (fixed seed). Paired deltas via bootstrap resampling on the same query set. Significance: 95% CI excludes zero.

4 Results

4.1 Per-Signal AUROC on MMLU-Pro

Log-probability is the best single signal on every model (Table 3). It significantly exceeds RouteLLM on Qwen ($\Delta = +0.049$, CI $[+0.008, +0.087]$) and ties on Llama and Mistral. Local accuracy: Qwen 43.7%, Llama 36.2%, Mistral 30.5%.

4.2 Cross-Dataset Transfer

RouteLLM collapses from 0.662 to chance (0.546) on TriviaQA (Table 4). Log-probability improves to 0.782. Per-model TriviaQA logprob: Qwen 0.828, Llama 0.800, Mistral 0.717.

4.3 Supervised Learning Curve

RouteLLM is unstable at small N (swings of 0.15–0.20), converges around $N=250$ –500, and at convergence never consistently exceeds the zero-shot line (Table 5, Figure 2).

4.4 Signal Latency

Log-probability requires 1.5–4.8s of local generation before routing (Table 6). A two-stage system can use GSA as a pre-generation filter (~ 500 ms) and logprob as a post-generation quality check.

4.5 Naive Fusion Hurts

Mean fusion drags the strong signal toward weaker ones (Table 7). The effect is largest on Qwen where the quality gap is widest.

4.6 Retrieval-Conditional Self-Assessment

Strong-hit-only retrieval ($\tau=0.70$) improves the signal by $+0.037$ (Table 8); including marginal hits ($\tau=0.60$) degrades it below the bare baseline. The design principle—never mention retrieval absence—is model-agnostic but density-dependent.

Table 3: AUROC on MMLU-Pro (95% bootstrap CI in brackets). Best zero-shot signal in bold. $n \approx 950$ per model.

Signal	Qwen-2.5-7B	Llama-3.1-8B	Mistral-7B
logprob	0.714 [.683,.746]	0.650 [.616,.687]	0.678 [.644,.716]
GSA v3	0.562 [.522,.598]	0.614 [.577,.652]	0.638 [.601,.674]
SC	0.504 [.490,.517]	0.594 [.558,.627]	0.604 [.569,.638]
KS	0.426 [.389,.465]	0.413 [.379,.452]	0.422 [.387,.459]
QC	0.524 [.495,.558]	0.521 [.491,.552]	0.522 [.491,.553]
RouteLLM-nm	0.664 [.629,.699]	0.662 [.628,.701]	0.676 [.641,.710]
RouteLLM-pks	0.665 [.629,.700]	0.644 [.609,.683]	0.676 [.641,.711]

Table 4: Cross-dataset AUROC (averaged across 3 models).

Signal	MMLU-Pro	TriviaQA	Δ
logprob	0.681	0.782	+0.101
GSA	0.605	0.703	+0.098
RouteLLM	0.662	0.546	− 0.116

5 Analysis

5.1 Why Does Log-Probability Work?

Average token log-probability measures how surprised the model was by its own output. On factual QA, a model that “knows” the answer produces high-probability tokens; a model that is guessing produces lower-probability, higher-variance tokens [Kadavath et al., 2022].

Models with calibration-aware RLHF produce stronger logprob signals (Table 9). Models without benefit more from explicit self-assessment. The two signals are complementary across the model-family axis.

5.2 Why Does Supervised Routing Collapse Cross-Dataset?

RouteLLM learns which regions of embedding space correspond to easy queries on MMLU-Pro. This transfers across model families (Qwen-trained RouteLLM achieves 0.663–0.675 on Llama/Mistral) because query difficulty is partially model-agnostic. But the embedding structure of MMLU-Pro is specific to MMLU-Pro. TriviaQA occupies a different region with different difficulty patterns.

Log-probability measures a property of the model’s generation, not of the query. It transfers because it does not depend on the query distribution.

5.3 Why Does Fusion Hurt?

Equal-weight fusion drags a strong signal toward weaker ones—a known failure mode when component quality is heterogeneous [Dietterich, 2000]. Thompson Sampling fusion, which should learn to upweight strong signals, collapses to equal-weight in our setup because no routing decisions are made during evaluation. Adaptive fusion with real feedback remains open.

5.4 Knowledge Similarity: A Same-Dataset Confound

KS is consistently inverted (AUROC 0.413–0.426). Per-category analysis reveals the mechanism: harder MMLU-Pro categories (law at 16.2% accuracy, engineering at 29.2%) have *higher* retrieval similarity because the KB was seeded from the same distribution. A KB seeded from external sources would produce a different distribution. We did not test this.

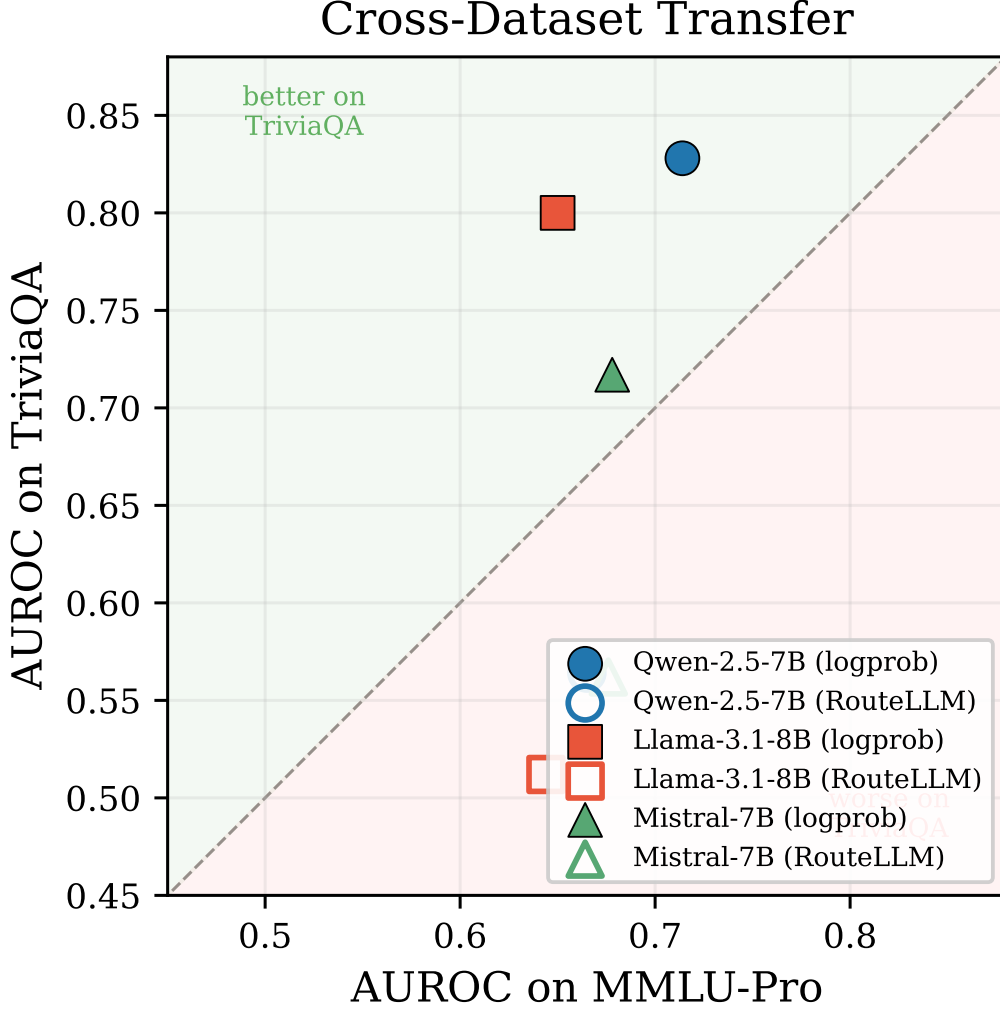


Figure 1: Cross-dataset transfer. Filled markers: logprob (above diagonal = better on TriviaQA). Open markers: RouteLLM (below diagonal = collapses on TriviaQA).

6 Cold-Start Deployment Guide

Recommendation. Use logprob from the first query (Table 10). If latency is critical, add GSA as a pre-generation filter. RouteLLM training is only justified when (a) the query distribution is known and stable, (b) pre-generation routing is required, and (c) cross-dataset transfer is not needed.

7 Related Work

Supervised routing. RouteLLM [Ong et al., 2024] trains classifiers on preference data. Extensions include contextual bandits [Chen et al., 2025], bandit feedback [Soare et al., 2025], dueling feedback [Li et al., 2025], non-stationary pricing [Taberner-Miller et al., 2026], and routing strategy surveys [Gomes et al., 2025]. All require training data. None benchmark against zero-shot alternatives.

Uncertainty-based routing. Chuang et al. [2025] benchmark uncertainty-based SLM routing across 1,500+ settings. Zhang et al. [2025] propose confidence-driven edge-cloud routing. Mukkunoth et al. [2025] combine three signals for pre-generation hallucination routing. Our work differs in directly comparing against supervised baselines and testing cross-dataset transfer.

Activation probes for success prediction. Most directly related to our work, Lugoloobi et al. [2026] train linear probes on pre-generation hidden-state activations to predict whether a model will

Table 5: RouteLLM-nm AUROC at increasing training sizes. Last row: logprob (zero-shot).

Training N	Qwen	Llama	Mistral
25	0.757	0.624	0.484
50	0.511	0.470	0.624
100	0.609	0.692	0.671
250	0.654	0.681	0.566
500	0.653	0.623	0.650
~ 1000	0.620	0.646	0.649
logprob (0)	0.714	0.650	0.678

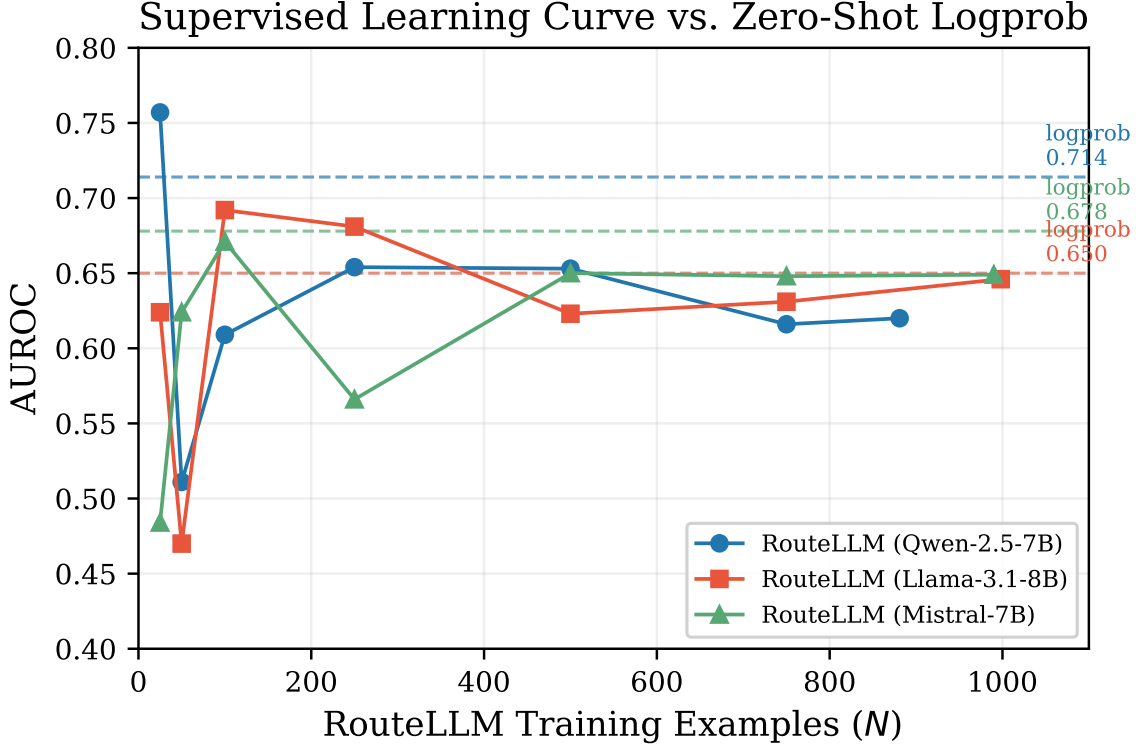


Figure 2: RouteLLM learning curve (solid) vs. logprob zero-shot AUROC (dashed), per model.

answer correctly, achieving AUROC > 0.7 on math and coding tasks. They demonstrate probe-guided routing with up to 70% cost reduction on MATH, and find that routing effectiveness is limited by probe reliability rather than routing sophistication—a conclusion we reach independently from a different direction. Critically, their approach requires access to model internals (residual-stream activations), limiting it to open-weight models with controlled inference. Our logprob signal requires only token probabilities available from any inference API, works zero-shot across model families without per-model probe training, and—as we show—transfers across datasets where supervised approaches fail. They do not compare against RouteLLM-style supervised baselines or test cross-dataset generalization.

Token-level calibration. Kadavath et al. [2022] showed calibrated token probabilities in RLHF’d models. Guo et al. [2017] established calibration theory. Huang et al. [2024] confirmed log-probability reliability in instruction-tuned models. We apply the same mechanism to routing and show it transfers where supervised approaches fail.

Self-assessment. Kadavath et al. [2022] tested $P(\text{True})$. Ren et al. [2023] reformulated generation into self-evaluation. Chen et al. [2024] found self-reported confidence outperforms self-consistency at lower cost. We extend self-assessment with retrieval-conditional prompting and show effectiveness is model-specific.

Table 6: Mean signal latency (ms). “Pre-gen” = routes before full local generation.

Signal	Qwen	Llama	Mistral	Pre-gen?
RouteLLM	<1	<1	<1	✓
KS (FAISS)	72	72	77	✓
GSA v3	461	777	767	✓
logprob	1,531	4,756	3,873	×
SC	539	2,874	2,766	×

Table 7: AUROC of signal combinations (simple mean) vs. logprob alone.

Combination	Qwen	Llama	Mistral
logprob alone	0.714	0.650	0.678
all 6 (mean)	0.523	0.643	0.643
logprob + GSA	0.634	0.645	0.669

Multi-signal uncertainty. UQLM [Lin et al., 2025] and MUSE [Geng et al., 2025] combine signals into unified frameworks. We find naive fusion hurts—consistent with ensemble theory [Dietterich, 2000] under heterogeneous quality, but not previously shown for LLM routing.

Self-consistency. Wang et al. [2022] proposed multi-path agreement. Our two-sample variant is model-specific: effective on Llama/Mistral, dead on Qwen.

Semantic entropy. Kuhn et al. [2023] proposed detecting confabulations via entropy over semantic clusters, published in *Nature*. Farquhar et al. [2024] extended this with cheaper approximations. These methods require 5–10 generations per query, making them 5–10 $\times$ more expensive than our single-generation logprob signal.

Verbalized confidence. Tian et al. [2023] showed LLMs can express calibrated confidence when prompted. Xiong et al. [2024] found verbalized confidence is often overconfident but improvable. Our GSA signal is a constrained variant that extracts YES-token probability rather than a numeric score, avoiding the overconfidence problem.

Uncertainty surveys. Geng et al. [2024] and Huang et al. [2025] provide comprehensive surveys. Our work is narrower but deeper: fewer methods, more models and datasets, with a supervised baseline comparison that surveys do not include.

8 Limitations

1. **Two datasets, one task type.** Both MMLU-Pro and TriviaQA are factual QA. Reasoning, creative, and multi-turn tasks are untested.
2. **Three models, one size.** All 7–8B. The calibration hypothesis predicts logprob improves with scale; untested.
3. **Post-generation routing cost.** Logprob requires full generation (1.5–4.8s) before routing. Wasted on escalated queries.
4. **Static KB.** All experiments use a frozen knowledge base. In production, the KB grows from escalations, making routing non-stationary.
5. **MCQ inflation.** MMLU-Pro’s 10-option format gives a 10% chance floor. TriviaQA partially addresses this.
6. **Label noise.** 13% inter-judge disagreement. Absolute AUROC carries $\sim\pm 0.05$ uncertainty; differential comparisons are robust.

Table 8: GSA v3 (retrieval-conditional) vs. v2 (bare) on Qwen MMLU-Pro ($n=931$).

Variant	AUROC	Queries w/ retrieval
GSA v2 (bare)	0.562	0 / 931
GSA v3 ($\tau=0.70$)	0.599	290 / 931 (31%)
GSA v3 ($\tau=0.60$)	0.511	769 / 931 (83%)

Table 9: Signal quality vs. RLHF emphasis.

Model	RLHF emphasis	logprob	GSA
Qwen-2.5-7B	Calibration-aware	0.714	0.562
Mistral-7B	Mixed	0.678	0.638
Llama-3.1-8B	Helpfulness	0.650	0.614

9 Conclusion

Training data is not necessary for effective LLM confidence estimation. Average token log-probability—a zero-shot signal available from the first query at zero cost—matches supervised baselines in-distribution and substantially outperforms them out-of-distribution. For practitioners deploying local-to-cloud routing: start with log-probability, skip the training, and invest engineering effort elsewhere.

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Table 10: Routing options at deployment time.

Stage	Signal	AUROC range	Cost	Latency
Day 0	logprob	0.650–0.833	\$0	1.5–4.8s
Day 0	GSA v3	0.562–0.720	\$0	0.5–0.8s
After labeling	RouteLLM	0.620–0.676	~\$25	<1ms

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